Hướng dẫn chạy tool LibreLane cho project I2C Master Controller

Link tool open-source chi tiết: https://librelane.readthedocs.io/en/latest/index.html

1. **Install Nix and clone LibreLane**

   There are a number of ways to install LibreLane on your Windows, Mac, or Linux computer. In this project, we will use the **Nix-based installation**, which is best system built for Linux and macOS and it is also the primary build system for LibreLane.

   Follow the instruction link below that suits the best for your operating system (Windows 10+ / macOS 14+ / Ubuntu / Other Linux)

   https://librelane.readthedocs.io/en/latest/installation/nix_installation/index.html

   After installing Nix and cloning LibreLane, turn to step 2.

2. **Install Nix packages**

   - Invoke nix-shell using command below:

   ```
   $ nix-shell ~/librelane/shell.nix
   ```

   - some packages will be downloaded (about 3GB) and then the terminal prompt should change to

   ```
   [nix-shell:~/librelane]$
   ```

   - You can test the tool by running the smoke test with following command:

   ```
   [nix-shell:~]$ librelane --log-level ERROR --condensed --show-progress-bar --smoke-test
   ```

3. **Running the flow**

   - Create a directory to add our source files and config.json file to:

   ```
   [nix-shell:~]$ mkdir -p ~/my_design/i2c
   ```

   - Copy the **source** folder and the **config.json** file into the directory.
   - Run the following command:

   ```
   [nix-shell:~]$ librelane ~/my_design/i2c/config.json
   ```

   - The tool will automatically run all the steps needed (about 5 minutes), move to step 4 after finish.

4. **Checking the results**

   - To view the layout (the final GDSII layout) run this command:

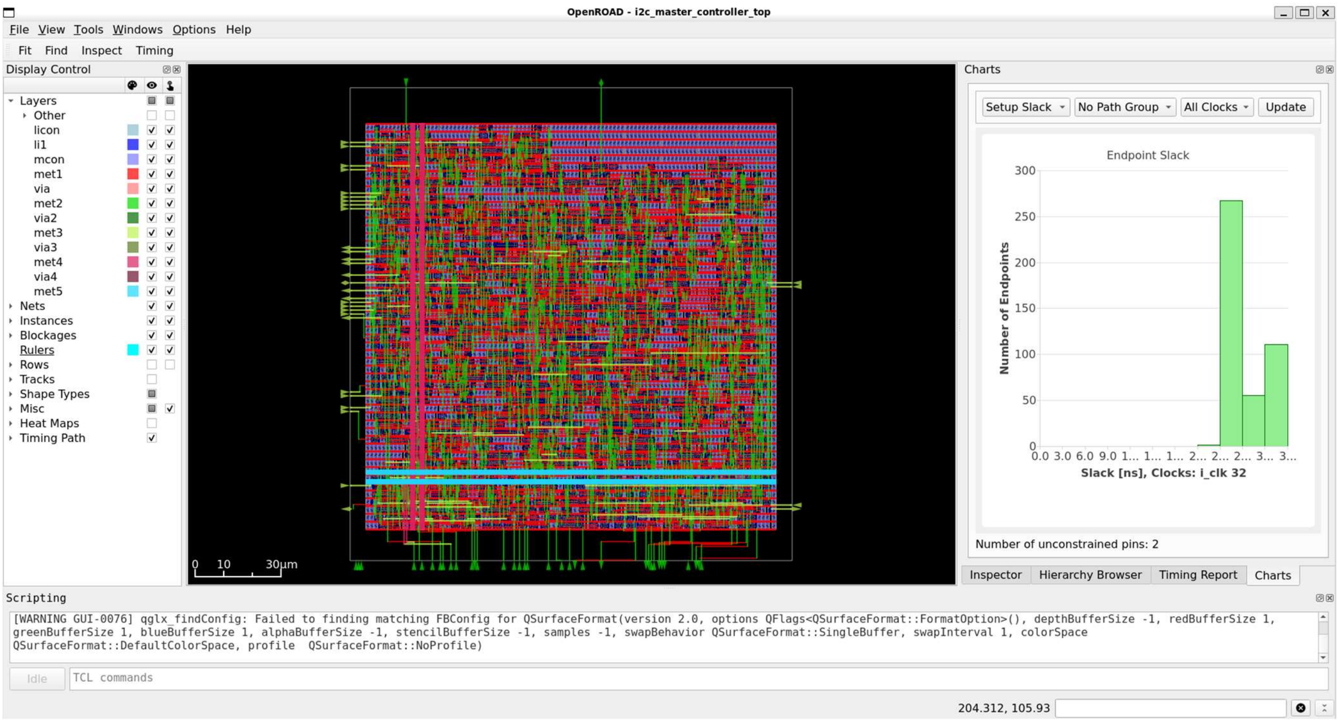
   ```
   [nix-shell:~]$ librelane --last-run --flow openinklayout ~/my_design/i2c/config.json
   ```

   - The Klayout app will open and you should see the following:

- Or if you want to view the layout in the **OpenROAD GUI**, use this command:

```
[nix-shell:~]$ librelane --last-run --flow openinopenroad ~/i2c/i2c_master/config.json
```

- And the app will show like below:



5. **Logs and reports directory**
   - You will find that a **runs directory** is created with timestamp (for example **runs/RUN_2026-04-13_16-59-15**). By default, LibreLane runs a Flow composed of a sequence of Step(s). Each step has its separate directory within the run directory.
   - Each step directory contains log files, report files, metrics and output artifacts created by the step. For example, the contents inside 70-netgen-lvs folder are:

```
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41]$ cd 70-netgen-lvs/
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41/70-netgen-lvs]$ ls
COMMANDS  config.json     netgen-lvs.log                      reports      state_in.json
_env.tcl  lvs_script.lvs  netgen-lvs.process_stats.json  runtime.txt  state_out.json
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41/70-netgen-lvs]$ |
```

- A **run directory** is composed of many of these step directories:

```
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41]$ ls
01-verilator-lint                      44-openroad-detailedrouting
02-checker-linttimingconstructs        45-odb-removeroutingobstructions
03-checker-linterrors                  46-openroad-checkantennas-1
04-checker-lintwarnings                47-checker-trdrc
05-yosys-jsonheader                    48-odb-reportdisconnectedpins
06-yosys-synthesis                     49-checker-disconnectedpins
07-checker-yosysunmappedcells          50-odb-reportwirelength
08-checker-yosyssynthchecks            51-checker-wirelength
09-checker-netlistassignstatements     52-openroad-fillinsertion
10-openroad-checksdcfiles              53-odb-cellfrequencytables
11-openroad-checkmacroinstances        54-openroad-rcx
12-openroad-staprepnr                  55-openroad-stapostpnr
13-openroad-floorplan                  56-openroad-irdropreport
14-openroad-dumprcvalues               57-magic-streamout
15-odb-checkmacroantennaproperties     58-klayout-streamout
16-odb-setpowerconnections             59-klayout-render
17-odb-manualmacroplacement            60-magic-writelef
18-openroad-cutrows                    61-odb-checkdesignantennaproperties
19-openroad-tapendcapinsertion         62-klayout-xor
20-odb-addpdnobstructions              63-checker-xor
21-openroad-generatepdn                64-magic-drc
22-odb-removepdnobstructions           65-klayout-drc
23-odb-addroutingobstructions          66-checker-magicdrc
24-openroad-globalplacementskipio      67-checker-klayoutdrc
25-openroad-ioplacement                68-magic-spiceextraction
26-odb-customioplacement               69-checker-illegaloverlap
27-odb-applydeftemplate                70-netgen-lvs
28-openroad-globalplacement            71-checker-lvs
29-odb-writeverilogheader              72-checker-setupviolations
30-checker-powergridviolations         73-checker-holdviolations
31-openroad-stamidpnr                  74-checker-maxslewviolations
32-openroad-repairdesignpostgpl        75-checker-maxcapviolations
33-odb-manualglobalplacement           76-misc-reportmanufacturability
34-openroad-detailedplacement          77-klayout-opengui
35-openroad-cts                        78-klayout-opengui
36-openroad-stamidpnr-1                79-openroad-opengui
37-openroad-resizertimingpostcts       error.log
38-openroad-stamidpnr-2                final
39-openroad-globalrouting              flow.log
40-openroad-checkantennas              resolved.json
41-odb-diodesonports                   tmp
42-openroad-repairantennas             warning.log
43-openroad-stamidpnr-3
```

- There is also a **final folder** containing several directories that contain all the different layout views produced by the flow. It looks like this:

```
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41/final]$ ls
def  json_h       lef  mag      metrics.csv   nl   pnl     sdc  spef   vh
gds  klayout_gds  lib  mag_gds  metrics.json  odb  render  sdf  spice
[nix-shell:~/i2c/i2c_master/runs/RUN_2026-04-15_09-27-41/final]$ 
```